

Unit 11, Lesson 5: Interior and Exterior Angles of Triangles



Benchmark

MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles.



Learning Targets

- ☐ I can show that the sum of the interior angles of a triangle is 180 degrees.
- ☐ I can show that the sum of the exterior angles of a triangle is 360 degrees.
- ☐ I can solve problems involving postulates, relationships, and theorems of triangles.



11.5.1 Warm-Up: Superhero

1. If geometry had a superpower, it would be ...

because ...

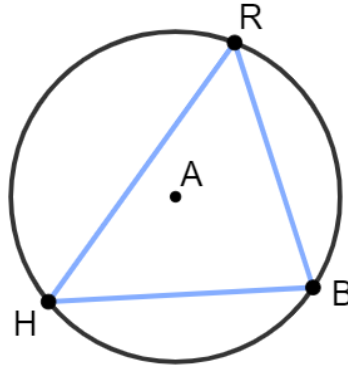
2. Geometry's villain would be ...

because ...



11.5.2 Exploration: Interior Angles of Triangles

1. Circle A is shown, where $m\widehat{RB} = 102^\circ$, $m\widehat{BH} = 108^\circ$, and $m\widehat{HR} = 150^\circ$.

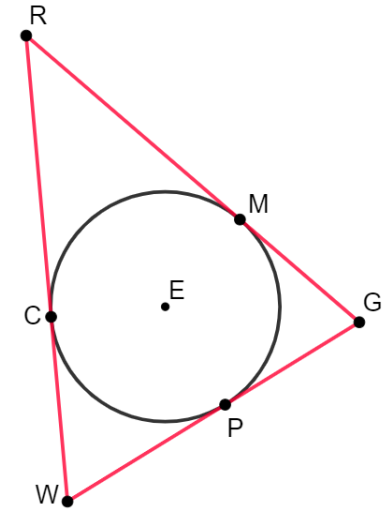


- a. Complete the table.

Angle Measures
$m\angle RHB =$
$m\angle RBH =$
$m\angle HRB =$

- b. What is the sum of the angles in $\triangle RHB$?

2. Circle E is shown where points P , C , and M are on the circle. $\overline{RG}$ and $\overline{RW}$ are tangent to Circle E and intersect at point R outside the circle. $\overline{GR}$ and $\overline{GW}$ are tangent to Circle E and intersect at point G outside the circle. $\overline{WR}$ and $\overline{WP}$ are tangent to Circle E and intersect at point W outside the circle. $m\widehat{CM} = 136^\circ$, $m\widehat{MP} = 108^\circ$, and $m\widehat{CP} = 116^\circ$.



- a. Complete the table.

Angle Measures
$m\angle WRG =$
$m\angle RGW =$
$m\angle GWR =$

- b. What is the sum of the angles in $\triangle RGP$?
- c. Write a conjecture about the sum of the interior angles of a triangle.

- d. Ask two classmates for their conjectures. Record their conjectures and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conjecture	My Summary of Their Reason	Initials

- e. Review your conjecture with your partner. If necessary, make any changes.



Triangle Sum Theorem

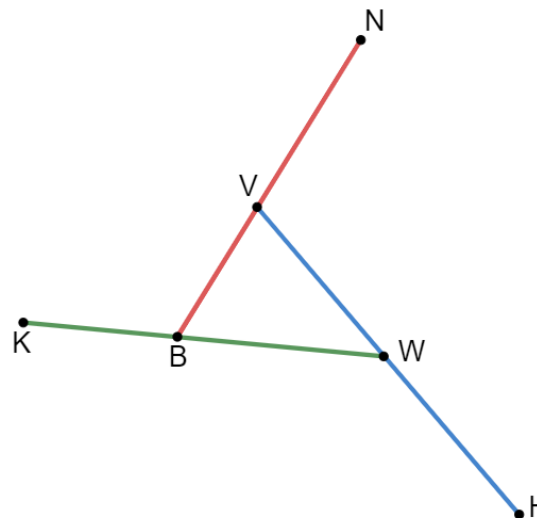
The sum of the measures of the interior angles of a triangle is 180° .



11.5.3 Activity: Exterior Angles of Triangles

Work with your partner to complete the following.

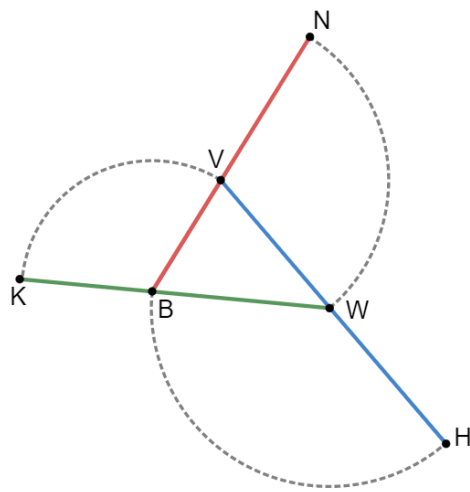
1. On a separate sheet of paper, draw three line segments, $\overline{BN}$, $\overline{VH}$, and $\overline{KW}$, to make $\triangle VBW$ as shown.



An **exterior angle** is an angle created by extending one leg of a triangle.

- List the exterior angles of $\triangle VBW$.
- Estimate the sum of the exterior angles of $\triangle VBW$.

2. Use a compass to draw arc $\widehat{BH}$ using point W as the center, $\widehat{KV}$ using point B as the center, and $\widehat{NW}$ using point V as the center, as shown.



- Trace the figure above onto patty paper and cut out each sector. Rearrange the sectors to determine the sum of the exterior angles.
- What do you notice about the sum of the exterior angles of the triangle?
- Write a conjecture about the sum of the exterior angles of a triangle.

- Ask two classmates for their conjectures. Record their conjectures and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

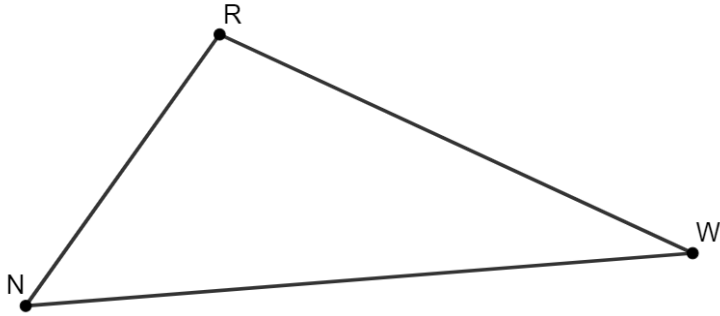
Conjecture	My Summary of Their Reason	Initials

- Review your conjecture with your partner. If necessary, make any changes.



11.5.4 Guided Practice: Interior and Exterior Angles of Triangles

1. $\triangle RWN$ is shown where $m\angle NRW = 14x + 3$, $m\angle RWN = \frac{9}{2}x - \frac{5}{2}$, and $m\angle WNR = 9x - 13$.

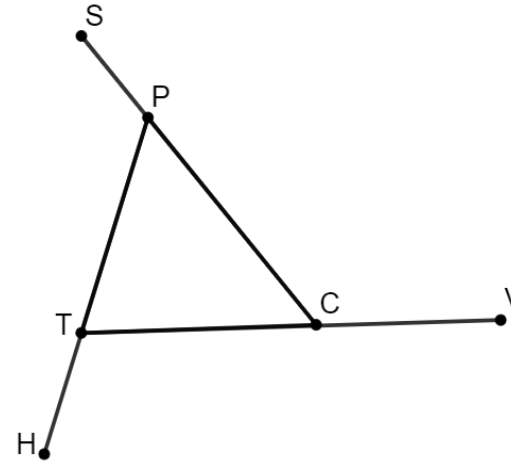


- a. Find the value of x .

- b. Complete the table.

$m\angle NRW =$	$m\angle RWN =$	$m\angle WNR =$
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2. $\triangle PCT$ is shown, where $\angle SPT$, $\angle PCV$, and $\angle HTC$ are exterior angles. $m\angle PCT = 52^\circ$, $m\angle SPT = (11x - 20)^\circ$, and $m\angle HTC = (8x + 5)^\circ$.



- a. Find the value of x .

- b. What is the measure of $\angle SPT$?

- c. What is the measure of $\angle HTC$?



11.5.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.



Benchmarks

MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Targets

- ☐ I can determine if a set of triangle side lengths can be used to construct a triangle.
- ☐ I can use the triangle inequality theorem to prove relationships in triangles.